

# Trustworthiness of Models based on Language

Responsible AI Project Report

Hannah Buch   Yassine Ajoud

## Abstract

We examine how reliable a large language model recognizes books across five languages. In each trial the model is given a short passage from a book and has to select the correct title from four candidates. All passages were taken from the Gutenberg project and the model is Qwen2.5-7B-Instruct. It reached 88% accuracy on 100 English books, 83% on Dutch, 72% on German, 71% on French and 66% on Italian. Alongside these five runs we tested German translations of English books, passages corrupted with random character noise and a variant in which the model has to justify its choice. English stays ahead of the other languages in every test. This means the model is better at recognizing books in English, so a user who works with it in another language gets weaker results.

## 1 Introduction

Large language models answer questions in many languages, yet most of their training data is in English. It is therefore reasonable to ask whether they perform as good in German or Italian as they do in English. The question matters in regard of trust. When a user in Germany addresses the model in German and the model knows less in that language, the answers are worse and the user may not notice.

The suggested course project approach was to take an existing benchmark, translate it with a chatbot and compare the scores before and after. We slightly departed from this plan for one reason. Machine translation introduces errors of its own, so when a score drops it becomes difficult to tell whether the tested model is weak in that language or whether the translated benchmark is simply broken. Books usually avoid the problem. The Gutenberg project hosts old books in many languages. The German, French, Italian and Dutch texts there were written by people rather than produced by a translation system. This lets us test the model in each language without any machine translation in between.

The test itself is straightforward. The model receives a passage of 1000 characters taken from the middle of a book together with four candidate titles, one of which is correct and it answers with the letter of the option it believes to be right. A model that encountered the book during training should do this well, while a model that never saw the book can still guess from the names and places in the passage. Random guessing would yield 25%.

We ran this test with Qwen2.5-7B-Instruct on 100 books each in English, German, French, Italian and Dutch. On top of these five runs we carried out three smaller experiments. In the first we took 23 German translations of English books and asked the model for the English original title. In the second we replaced half of the characters in the passages with random letters and observed the effect on accuracy. In the third we required the model to explain its answer on a small set of English books.

## 2 Related Work

Chang et al. investigated which books ChatGPT and GPT-4 have seen [1]. They took short passages from novels, masked a single character name and let the model fill in the missing name. When a model supplies the correct name repeatedly it has most likely encountered the book during training. They found that the models have memorized a large number of books and that the books whose passages appear most often on the web are memorized best. Our setup asks a similar question with a simpler tool. Rather than asking for names inside the text we ask for the title among four options.

Zhang et al. studied the gap between English and other languages in large language models [2]. They gave the same tasks to the models in several languages and compared the results. For some tasks the correct answer stays the same after translation, while for others it depends on the language itself. On these language-dependent tasks the models had clear difficulty once the question was no longer in English. The title of their paper states the conclusion directly. Do not trust ChatGPT when your question is not in English. Our results point in the same direction, since the recognition task yields weaker scores in every language we tried other than English.

## 3 Approach

### 3.1 Data

The books come from the dataset `manu/project_gutenberg` on Hugging Face, which holds the raw text of Project Gutenberg books. We stream the dataset so that the code reads one book at a time over the network and stops once it has enough, which avoids downloading the whole collection.

Every raw file begins with a header that holds lines like Title, Author and sometimes Translator. We read the title from the header and convert it to lowercase. Books without a title line are skipped, as are titles we have already seen. When the header contains a Translator line we treat the book as a translation and drop it, which keeps the native sets clean. For each accepted book we cut a passage of 1000 characters from the middle of the file. The middle is a safe choice because the start of a Gutenberg file holds license text, a table of contents and often a preface, whereas the middle is almost always story text. In this way we collect 100 books per language.

### 3.2 Questions

Every book becomes a single multiple choice question that holds the passage and four titles, exactly one of which is correct. The three wrong titles are drawn at random from the other 99 books of the same language set. The four options are then shuffled and labelled A to D. Because the wrong options are real titles in the same language, the model can not come to the answer by language alone.

### 3.3 Model and prompt

The model is Qwen2.5-7B-Instruct [?], an open instruction tuned model with 7 billion parameters that we run on a local machine through the transformers library. The prompt states that the task is to identify the book behind a passage, then presents the passage and the four

options and asks for a single letter. We allow the model to generate at most 4 tokens with greedy decoding so that the answer stays short and stable. A regular expression extracts the first letter from A to D in the reply and when no letter is found the question counts as an error. Across the five base runs and the translation run this case never came up as every answer contained a readable letter.

### 3.4 German translations of English books

This set is harder, because the passage is in German while the answer options are the English original titles. To get it right the model has to connect the content of a book across two languages.

Assembling the set took two steps. A script first searched the German split for signs of a translation from English. It matched phrases like *aus dem Englischen* in the opening part of the file and also flagged books that carry a translator entry together with an author from a list of well known English writers. The script wrote every hit to a CSV file, after which we filled in the English original titles by hand. For some hits no clear original could be established, so the final list holds 23 usable books, shown in Table 1. Eleven of the 23 files are volumes of a single work. For this set the passage is 3000 characters long and starts 1000 characters into the file.

| Author                              | German title                                                                          | English original title                                | Files |
|-------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|-------|
| Thomas Babington Macaulay           | Geschichte von England seit der Thronbesteigung Jakob's des Zweiten (volumes 1 to 11) | The History of England from the Accession of James II | 11    |
| Oscar Wilde                         | Das Bildnis des Dorian Gray                                                           | The Picture of Dorian Gray                            | 1     |
| Charles Dickens                     | Zwei Städte                                                                           | A Tale of Two Cities                                  | 1     |
| Oscar Wilde                         | Salome. Tragödie in einem Akt                                                         | Salome                                                | 1     |
| William Shakespeare                 | Der Kaufmann von Venedig                                                              | The Merchant of Venice                                | 1     |
| Paschal Beverly Randolph            | Dhoula Bel. Ein Rosenkreuzer-Roman                                                    | Ravalette                                             | 1     |
| Charles Dickens                     | Der Weihnachtsabend. Eine Geistergeschichte                                           | A Christmas Carol                                     | 1     |
| Metta Victoria Fuller Victor        | Tagebuch eines bösen Buben                                                            | A Bad Boys Diary                                      | 1     |
| Daniel Defoe                        | Die Pest zu London                                                                    | A Journal of the Plague Year                          | 1     |
| Mark Twain                          | Die Abenteuer Tom Sawyers                                                             | The Adventures of Tom Sawyer                          | 1     |
| Harriet Beecher Stowe               | Onkel Tom's Hütte oder die Geschichte eines christlichen Sklaven, Band 1              | Uncle Tom's Cabin                                     | 1     |
| J. W. von Goethe and Thomas Carlyle | The Life of Friedrich Schiller, Introduction to German Translation                    | The Life of Friedrich Schiller                        | 1     |
| Lewis Carroll                       | Alice's Abenteuer im Wunderland                                                       | Alice's Adventures in Wonderland                      | 1     |

Table 1: The 23 German translations of English books in our test set.

### 3.5 Noise

A further test measures how much corrupted text the model can tolerate. A function passes over a passage and replaces a share of its characters with random letters from the ASCII alphabet. At 50% noise half of all characters are random. We applied this level to 20 German books with German title options.

### 3.6 Answers with explanations

In the base test the model prints a single letter and nothing else. We also wanted to observe how it reasons. In this variant the model must answer in a fixed format with the letter first and one or two sentences of explanation after it and the limit rises to 80 new tokens instead of 4. We ran the variant on the first 10 English questions.

## 4 Experiments

Table 2 lists all runs. Each run used the same model, the same greedy decoding and a single pass over its question set with no repetition. Every number in the next section therefore comes from one pass. The five base runs use 100 books per language, while the three additional runs are smaller.

| Run          | Text    | Titles  | Books | Snippet               |
|--------------|---------|---------|-------|-----------------------|
| English      | English | English | 100   | 1000 chars, middle    |
| German       | German  | German  | 100   | 1000 chars, middle    |
| French       | French  | French  | 100   | 1000 chars, middle    |
| Italian      | Italian | Italian | 100   | 1000 chars, middle    |
| Dutch        | Dutch   | Dutch   | 100   | 1000 chars, middle    |
| Translations | German  | English | 23    | 3000 chars, early     |
| Noise        | German  | German  | 20    | 1000 chars, 50% noise |
| Explanations | English | English | 10    | 1000 chars, middle    |

Table 2: Overview of all runs.

### 4.1 Results

Table 3 reports the outcome of all runs. Every base run stays well above the 25% guess line. English is the strongest set with 88 of 100 books, followed by Dutch with 83. German and French sit close together at 72 and 71, and Italian is the weakest at 66.

| Test set                    | Books | Correct | Accuracy |
|-----------------------------|-------|---------|----------|
| English                     | 100   | 88      | 88.0%    |
| Dutch                       | 100   | 83      | 83.0%    |
| German                      | 100   | 72      | 72.0%    |
| French                      | 100   | 71      | 71.0%    |
| Italian                     | 100   | 66      | 66.0%    |
| German text, English titles | 23    | 22      | 95.7%    |
| German with 50% noise       | 20    | 6       | 30.0%    |
| English with explanations   | 10    | 5       | 50.0%    |

Table 3: Accuracy of Qwen2.5-7B-Instruct on all runs. Random guessing gives 25%.

The translation run reached 22 of 23 books. The model read German text and still selected the correct English original title in almost every case.

The noise run fell sharply. Without noise the German base run stands at 72%, but once half of the characters are replaced the model got 6 of 20 books right. That is 30% against a guess line of 25%, so at this noise level the recognition ability is largely gone.

The explanation run reached 5 of 10 books where the base English run answered 6 of the same 10 questions correctly. The samples are tiny, so a difference of one book carries little weight. The content of the explanations is more revealing than the score. For book 1 the model chose the right option and named *The Tatler* as the clue in the text. For book 2 it chose a wrong option and wrote that the passage is clearly from *The Prodigals and their Inheritance* because the character Mr.Probert is specific to that book. Mr.Probert is in fact a character in *The Reverberator*, which was the correct answer. The model sounded equally certain in both cases.

## 5 Analysis

The plain reading of Table 3 is that the model recognizes literature best in English. German scores 16 points below English and French 17, while Italian is 22 points below. Dutch loses only 5 points, which is higher than we expected. Part of the reason may be a difference in filtering. The English and German runs removed translated books before building questions, whereas the French, Italian and Dutch runs kept them because the translation filter was not active there. A Dutch or French copy of a famous English book is easier to recognize than a local original so the gap for these three languages is probably somewhat larger than the table suggests.

The 95.7% for German translations with English titles looks very high next to the 72% for German books with German titles and we would not read too much into it. The set is small at 23 books and the books are famous with Dickens, Twain, Wilde, Shakespeare and Carroll all present. Eleven files are volumes of the same history so the same correct title recurs eleven times. The passages for this run were also three times longer and the answer pool was small. All of this pushes the score upward. The model can link German text to an English title when the book is well known, which fits the finding of Chang et al. [1]. Famous books have many copies on the web in several languages so the model had many opportunities to learn the link during training.

The noise run gives a clear picture. At 50% random characters the score lands at 30% against a guess line of 25%. A passage in which every second character is random is hard

for a human to read as well. A single noise level is of course a thin basis.

The explanation run is small yet it exposes a known problem from a new angle. The model states wrong facts with full confidence. The Mr.Probert case is a good example, as the explanation picks a real detail from the passage and attaches it to the wrong book. A user who does not know both books cannot catch the error. For trustworthiness this weighs more than the raw accuracy numbers because a wrong letter is cheap while a wrong letter wrapped in a fluent and confident account can mislead a reader.

Some limitations remain. Every run is a single pass so the numbers carry sampling luck and with 100 books a swing of a few points between repeated runs would be normal. The titles come from the file headers and were lowercased, so a few odd or broken titles may have entered the question sets. The three wrong titles per question are random draws, which makes some questions easy when the wrong titles fall far from the topic of the passage.

## 6 Conclusion

We built a small test around one clear question. Does a language model recognize books in other languages as well as it does in English. For Qwen2.5-7B-Instruct the answer is no. English reached 88% while the other four languages landed between 66% and 83%. The model beats random guessing everywhere, so the ability does not vanish outside English but it does weaken. The additional runs sharpen the picture. The model links German translations to their English originals when the books are famous, it breaks down when half of the text is random noise and its explanations sound convincing even when they are wrong.

## Author contribution statement

Both authors contributed to the project and discussed the design and code implementation. Hannah Buch was the main contributor in building and running the experiments. Yassine Ajoud focused on the written report and the presentation.

## GenAI use

DeepL was used for simple translations and for wording improvements in the written report.

## References

- [1] Kent K. Chang, Mackenzie Cramer, Sandeep Soni and David Bamman. Speak, Memory: An Archaeology of Books Known to ChatGPT/GPT-4. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.00118.
- [2] Xiang Zhang, Senyu Li, Bradley Hauer, Ning Shi and Grzegorz Kondrak. Don't Trust ChatGPT when Your Question is not in English: A Study of Multilingual Abilities and Types of LLMs. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.16339.